

A. PRODUCT IDENTITY

- **Product:** ARM & HAMMER Baking Soda
- **Compund:** Sodium bicarbonate ($NaHCO_3$)
- **Why it matters:** Widely used in baking as a leavening agent and has various practical household uses.
- **Student:** Jasmin Kate Brizuela, Chemistry for Engineers

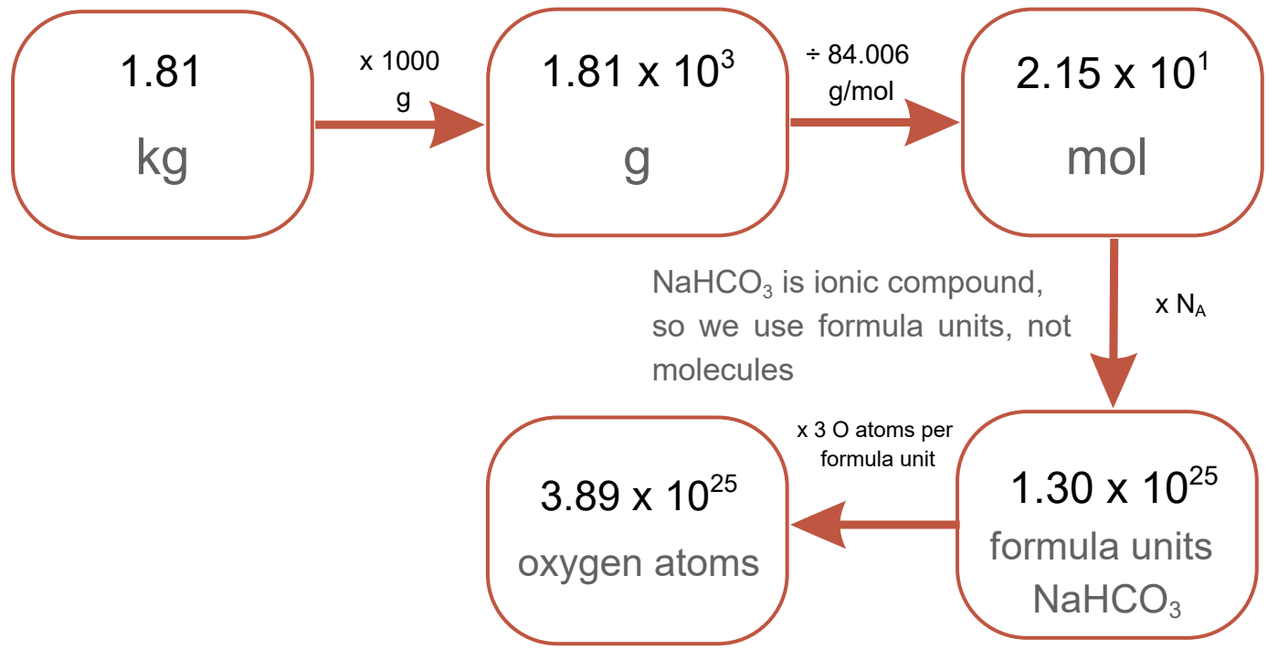
C. MOLAR MASS BUILD-UP

Element	Atoms	g/mol each	g/mol	% by mass
Na	1	22.990	22.990	27.365%
H	1	1.008	1.008	1.200%
C	1	12.011	12.011	14.297%
O	**3**	15.999	47.997	57.138%
Total	**6**	**-**	**84.006**	**100%**

B. BASE QUANTITATIVE DATA

Quantity	Value	Source	Precision
Net content	1.81 kg	Arm & Hammer Product label	product label; 3 s.f
Atomic mass Na, H, C, O	22.990, 1.008, 12.011, 15.999	CIAAW(2024)	not limiting
Avogadro Constant, N_A	$6.02214076 \times 10^{23}$ mol^{-1}.	BIPM(2019)	exact count
Formula $NaHCO_3$	subscripts 1, 1, 1, 3	PubChem	exact counts

D. THE MOLE MAP: 1.81 kg



E. MOLE-CONVERSION CALCULATIONS

1 Molar mass of $NaHCO_3$

22.990 + 1.008 + 12.011 + 3(15.999) = **84.006 g/mol**

2 Label Bridge

$$\cancel{1.81\ kg} \times \frac{1000\ g}{1\ \cancel{kg}} = 1810\ g \rightarrow \mathbf{1.81 \times 10^3\ g}$$

3 Mass to moles

$$\frac{1.81 \times 10^3\ \cancel{g}}{84.006\ \cancel{g}/mol} = 21.546\ mol \rightarrow \mathbf{2.15 \times 10^1\ mol\ NaHCO_3}$$

4 Moles to molecules

21.546 mol x $6.02214076 \times 10^{23}$ formula units

= **1.30×10^{25} formula units**

5 Atoms of one element inside the formula

1.30×10^{25} formula unit x 3 = **3.89×10^{25} oxygen atoms**

6 Percent Composition and two route check (Oxygen)

$$\frac{3(15.999)}{84.006} \times 100 = \mathbf{57.138\%\ oxygen}$$

Route A: 1810 g x 0.57138 $\approx$ **1034.19 g O**

Route B: 21.546 mol **$NaHCO_3$** $\times \frac{3\ mol\ O}{1\ mol\ \mathbf{NaHCO_3}} \times \frac{15.999\ g\ O}{1\ mol\ O}$

$\approx$ **1034.19 g O**

Both routes → **1.034×10^3 g O** ✓

7 From percent composition back to the formula

In 100 g: **Na** 27.365 g ÷ 22.990 = 1.190 mol **C** 14.297 g ÷ 12.011 = 1.190 mol

H 1.200 g ÷ 1.008 = 1.190 mol **O** 57.138 g ÷ 15.999 = 3.571 mol

÷ 1.190 = **1.000 : 1.000 : 1.000 : 3.000** → **$NaHCO_3$** (emperical=molecular, n=1)

8 Reverse: How many packages contain one mole?

$$1.00\ \cancel{mol} \times \frac{84.006\ \cancel{g}}{1\ \cancel{mol}} \times \frac{1\ package}{1810\ \cancel{g}} = 0.04641\ package \rightarrow \mathbf{4.64 \times 10^{-2}\ package}$$

One mole is approximately 4.64×10^{-2} package of a 1.81 kg package under the stated assumption.

F. SCALE-UP: FROM ONE PACKAGE TO A LARGER GROUP

1. 1 PACKAGE
1.81 kg
2.15×10^1 mol
1810 ÷ 84.006 g/mol

2. ONE GROUP
Assume 30 package
6.46×10^2 mol
30 x 21. 546 mol= 646.38 mol

3. CAMPUS
Assume 10 groups
6.46×10^3 mol
300 x 21. 546 mol= 6463.8 mol

4. REVERSE
One mole in 1 package is
4.64×10^{-2}
Approximately 84.0 g, or 4.64% of one package.

The 30 packages and 10 groups are illustrative assumptions, not actual usage figures.

G. SIGNIFICANT-FIGURE CHECK

Calculator: $1.294760263 \times 10^{25}$ formula units

Reported: 1.30×10^{25} formula units

- **Limits:** 1.81 kg (3 s.f). **Exact** N_A , subscript
- **Guard digits kept until the last calculations**

ONE NUMBER TAKEAWAY

"One 1.81 kg package of Arm & Hammer baking soda contains approximately 1.30×10^{25} formula units of $NaHCO_3$, showing how an ordinary household product can contain an incredibly large number of chemical particles."

AI was used to assist in organizing the Atom Count Card, explaining the mole conversions, and checking the calculations. The student is responsible for verifying the product label, references, and final answers.

H. REFERENCES (APA 7TH EDITION)

Church & Dwight Co., Inc. (n.d.). ARM & HAMMER pure baking soda value size

Commission on Isotopic Abundances and Atomic Weights. (n.d.). Standard atomic weights.https://ciaaw.org/atomic-weights.

National Center for Biotechnology Information. (n.d.). PubChem compound summary for CID 516892, sodium bicarbonate.PubChem. https://pubchem.ncbi.nlm.nih.gov/compound/516892

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